

Course InformationCourse Code: **CS-08519**

Credit Hours: 4 (3+1)

Lab Hours/Week: 3

Course Title: **Computer Networks**

Lecture Hours/Week: 3

Pre-Requisites: **None****i. Design Skills / Techniques Practiced**

The *Computer Networks (CS-08519)* course provides students with essential design-oriented skills through simulation-based and problem-solving activities. These design skills enable students to conceptualize, model, and implement functional network infrastructures that reflect real-world scenarios. The emphasis is on **hands-on configuration, topology optimization, and performance evaluation** using modern networking technologies.

Students learn to design and test secure, scalable, and efficient networks by applying theoretical principles in practical environments such as Cisco Packet Tracer, Wireshark, and Huawei eNSP. The course ensures that learners not only understand networking concepts but can **apply design thinking** to solve connectivity, routing, and communication challenges in dynamic organizational contexts.

Table 1: Design Skills and Techniques Practiced in Laboratory and Class Activities

Sr. No.	Design Skill / Technique	Description and Purpose
1	Network Topology Design	Designing LAN, WAN, and hybrid network layouts using Cisco Packet Tracer to understand inter-device connectivity, addressing schemes, and traffic flow.
2	IP Addressing and Subnetting	Applying subnetting techniques to allocate IP ranges efficiently, ensuring scalability and minimal wastage of addresses.
3	Routing Protocol Implementation	Designing and configuring routing protocols (RIP, OSPF, and BGP) to establish dynamic communication paths between networks.
4	Switching and VLAN Configuration	Creating and testing VLANs for network segmentation, performance enhancement, and improved security.
5	Wireless Network	Planning and simulating wireless LAN architectures with access

Sr. No.	Design Skill / Technique	Description and Purpose
	Design	points and SSID configurations for mobile connectivity.
6	Network Security and Monitoring	Using Wireshark to analyze traffic patterns, detect anomalies, and apply basic security configurations such as ACLs.
7	Performance Optimization	Evaluating throughput, latency, and bandwidth utilization to improve network reliability and response times.
8	Documentation and Troubleshooting	Preparing design documentation and troubleshooting reports to align with professional network engineering standards.

Outcome Alignment

Through these design-focused exercises, students gain **practical competencies in network architecture development, configuration, and optimization**. The practiced techniques directly support Course Learning Outcomes (CLO-3, CLO-4, and CLO-5), ensuring that learners are ready for both academic research and industry applications in areas such as **network design, system integration, and intelligent infrastructure management**.